Rules

Sunday, April 5, 2026 9:32 PM

$$\text{Mean} = \frac{x_1+x_2+x_3+x_4}{4}$$

$$\text{Median} = 1,3,5,7 \rightarrow \frac{3+5}{2} = 4$$

Mode = Most repeated value

range = Max − Min

$$\text{Variance} = \frac{\Sigma(x-\bar{x})^2}{n}$$

$$\text{Standard deviation} = \sqrt{\text{variance}}$$

$$\text{Skewness} = \frac{\Sigma(x-\bar{x})^3}{n s^3} \quad (s \rightarrow \sigma)$$

$$\text{Kurtosis} = \frac{\Sigma(x-\bar{x})^4}{n s^4}$$

$$IQR = Q_3 - Q_1$$

Lower / upper bounds

$Q_1 - 1.5 \times IQR$ $\quad$ $Q_3 + 1.5 IQR$

3 instead of 1.5

Extreme outliers:

Normal → Skewness = 0 Kurtosis = 3

$$\text{Covariance} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{n-1} \rightarrow = -, +, 0$$

$$\text{Corelation} = \frac{Cov(x,y)}{\sigma_x \sigma_y} \rightarrow -1 \leq x \leq 1$$

$$\text{Min-Max Scaling} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

→ affected by outliers

→ $0 \leq x \leq 1$

$$\text{Z-Score Scaling} = \frac{x - \bar{x}}{\sigma_x} \rightarrow -\infty \leq x < \infty$$

Z-Score Scaling = $\frac{X - \bar{X}}{\sigma_X}$ → $-\infty \leq x < \infty$

Robust Scaling = $\frac{(X - X_{med})}{IQR}$ → best for skewed data/outliers

decimal scaling = $\frac{X}{10^J}$ $J \rightarrow |X| < 1$

log transformation = log data →

Feature	Value
Feature1	10
Feature2	100
Feature3	1000
Feature4	10000

→

Feature	Value
Feature1	1.0
Feature2	2.0
Feature3	3.0
Feature4	4.0

useful for long-tailed data
only for nonzero/non-negative values

Box-Cox = turns the +ve values into a normal distribution
Johnson = " " but includes -ve data too